Lecture 7. Properties of Expectation

Linearity, covariance, conditional expectation, prediction, moment generating functions, multivariate normal

Contents

7.1 Expectation of sums **2**

7.2 Covariance and variance of sums **2**

7.3 Conditional expectation **3**

7.4 Conditional expectation and prediction **5**

7.5 Moment generating functions **5**

7.6 The multivariate normal distribution **6**

7.7 General definition of expectation **6**

7.8 Selected exercises **6**

7.1 Expectation of sums

$E[g(X,Y)] = \sum_x \sum_y g(x,y)p(x,y)$ or $\iint g(x,y)f(x,y)\,dx\,dy$ (as Proposition 5.2). With $g = x + y$: $E[\sum X_i] = \sum E[X_i]$, no independence needed. $X \geq Y$ implies $E[X] \geq E[Y]$.

Example 7.1

(a) Accident at $X \sim U(0,L)$, ambulance at independent $Y \sim U(0,L)$: $E|X-Y| = \frac{1}{L^2}\int_0^L \int_0^L |x-y|\,dx\,dy = \frac{L}{3}$.
(b) Boole: with X the number of A_i occurring and $Y = \mathbb{1}\{X \geq 1\} \leq X$, $P(\bigcup A_i) = E[Y] \leq \sum P(A_i)$.
(c) $E[\bar{X}] = \mu$ for identically distributed X_i.

Example 7.2 (Indicator calculations)

(a) Matches: $E[\#\text{matches}] = N \cdot \frac{1}{N} = 1$ for every N.
(b) Coupons: with i types held, the wait for a new one is geometric with parameter $(N-i)/N$, so

$$E[T] = \sum_{i=0}^{N-1} \frac{N}{N-i} = N\Big(1 + \frac{1}{2} + \cdots + \frac{1}{N}\Big) \approx N \log N.$$

Distinct types in n purchases: $N\{1-(1-1/N)^n\}$ in expectation.
(c) Runs: in a random arrangement of n ones and m zeros, with I_i indicating a run of ones starting at i, $P(I_1 = 1) = \frac{n}{n+m}$ and $P(I_i = 1) = \frac{mn}{(n+m)(n+m-1)}$ $(i > 1)$, so the expected number of runs is $\frac{n(m+1)}{n+m}$.
(d) Quicksort (random pivot, compare, recurse): the i-th and j-th smallest are compared iff one of them is the first pivot among the $j-i+1$ values between them, probability $2/(j-i+1)$:

$$E[\#\text{comparisons}] = \sum_{i<j} \frac{2}{j-i+1} = 2(n+1)\sum_{k=1}^{n} \frac{1}{k} - 4n \approx 2n \log n.$$

Proposition 7.1 (Moments of the number of events)

If X is the number of the events $A_1, \ldots, A_n$ that occur, then

$$E\Big[\binom{X}{k}\Big] = \sum_{i_1 < \cdots < i_k} P(A_{i_1} \cdots A_{i_k}).$$

Proof. $\binom{X}{k} = \sum_{i_1<\cdots<i_k} I_{i_1} \cdots I_{i_k}$. ∎

$k = 2$: $E[X^2] = E[X] + 2\sum_{i<j} P(A_iA_j)$. For matches $P(A_iA_j) = \frac{1}{N(N-1)}$, so $E[X(X-1)] = 1$ and $\mathrm{Var}(X) = 1$, as for Poisson(1).

7.2 Covariance and variance of sums

Motivation. The variance of a sum expands into covariances.

Proposition 7.2

If X, Y are independent, $E[g(X)h(Y)] = E[g(X)]E[h(Y)]$.

Proof. $\iint g(x)h(y)f_X(x)f_Y(y)\,dx\,dy$ factors. ■

$\mathrm{Cov}(X,Y) = E[XY] - E[X]E[Y]$. Independence implies Cov $= 0$, not conversely: X uniform on $\{-1,0,1\}$, $Y = X^2$ give Cov $= E[X^3] = 0$.

Proposition 7.3

Covariance is symmetric and bilinear: $\mathrm{Cov}(\sum_i a_iX_i, \sum_j b_jY_j) = \sum_{i,j} a_ib_j \mathrm{Cov}(X_i,Y_j)$, and $\mathrm{Cov}(X,X) = \mathrm{Var}(X)$. Hence

$$\mathrm{Var}\Big(\sum_i X_i\Big) = \sum_i \mathrm{Var}(X_i) + 2\sum_{i<j} \mathrm{Cov}(X_i,X_j),$$

which is $\sum \mathrm{Var}(X_i)$ for pairwise independent variables.

Example 7.3

(a) For i.i.d. X_i, $\mathrm{Var}(\bar X) = \sigma^2/n$, and $\sum(X_i - \bar X)^2 = \sum(X_i-\mu)^2 - n(\bar X - \mu)^2$ gives

$$E[S^2] = \frac{1}{n-1}(n\sigma^2 - n\cdot\sigma^2/n) = \sigma^2, \qquad S^2 = \frac{1}{n-1}\sum(X_i-\bar X)^2.$$

(b) Hypergeometric: for draw indicators, $\mathrm{Cov}(X_i,X_j) = \frac{m(m-1)}{N(N-1)} - \frac{m^2}{N^2} = -\frac{m(N-m)}{N^2(N-1)}$, giving $\mathrm{Var}(X) = n\frac{m}{N}(1-\frac{m}{N})\frac{N-n}{N-1}$.
(c) Multinomial: $N_i = \sum_k I_i(k)$ with $I_i(k)I_j(k) = 0$ and independent trials, so $\mathrm{Cov}(N_i,N_j) = n(0 - p_ip_j) = -np_ip_j$.

The **correlation** is $\rho(X,Y) = \mathrm{Cov}(X,Y)/\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}$.

Proposition 7.4

$-1 \le \rho(X,Y) \le 1$, with $\rho = \pm 1$ iff $Y = a + bX$ with probability 1 for some b of the same sign.

Proof. $0 \le \mathrm{Var}(X/\sigma_X \pm Y/\sigma_Y) = 2(1 \pm \rho)$, with equality iff $X/\sigma_X \pm Y/\sigma_Y$ is constant. ■

ρ measures *linear* dependence; for the bivariate normal (Example 6.11) $\rho(X,Y) = \rho$.

7.3 Conditional expectation

Motivation. $E[X \mid Y]$, the best guess of X given Y, averages back to $E[X]$: compute expectations by conditioning.

$E[X \mid Y = y] = \sum_x x p_{X|Y}(x \mid y)$ or $\int x f_{X|Y}(x \mid y)\,dx$; $E[X \mid Y] = g(Y)$ with $g(y) = E[X \mid Y = y]$.

Theorem 7.5 (Tower property)

$E[X] = E[E[X \mid Y]]$, i.e. $E[X] = \sum_y E[X \mid Y = y]P(Y = y)$ or $\int E[X \mid Y = y]f_Y(y)\,dy$.

Proof. Discrete case: $\sum_y \sum_x x p_{X|Y}(x \mid y) p_Y(y) = \sum_x \sum_y x p(x,y) = E[X]$. ■

Example 7.4 (The trapped miner)

A miner picks one of three doors at random: safety in 3 hours, or back to the mine after 5 or 7 hours. Conditioning on the first door, $E[X] = \frac{1}{3}\{3 + (5 + E[X]) + (7 + E[X])\}$, so $E[X] = 15$.

Example 7.5 (Random sums)

If $N \geq 0$ is integer-valued, independent of i.i.d. X_i, then $E[\sum_{i=1}^{N} X_i \mid N = n] = nE[X]$, so

$$E\Big[\sum_{i=1}^{N} X_i\Big] = E[N]E[X].$$

Example 7.6 (Waiting for k consecutive successes)

N_k: trials until k consecutive successes. After N_{k-1}, one more trial succeeds or restarts: $E[N_k \mid N_{k-1}] = N_{k-1} + 1 + (1-p)E[N_k]$, giving $E[N_k] = \frac{1}{p} + \frac{E[N_{k-1}]}{p}$ and

$$E[N_k] = \frac{1}{p} + \frac{1}{p^2} + \cdots + \frac{1}{p^k}.$$

For a fair coin, $E[N_2] = 6$ and $E[N_3] = 14$.

Probabilities by conditioning. $P(E) = E[P(E \mid Y)]$; e.g. $P(X < Y) = \int F_X(y) f_Y(y)\, dy$ for independent continuous X, Y, and $P(X + Y < a) = \int F_X(a - y) f_Y(y)\, dy$.

Example 7.7 (The best-prize problem)

n distinct prizes appear in random order; each is accepted or rejected on the spot, knowing only relative ranks. Reject the first k, then accept the first prize better than all earlier. If the best is at $X = i > k$, we win iff the best of the first $i - 1$ is among the first k, probability $k/(i-1)$:

$$P_k(\text{best}) = \frac{1}{n} \sum_{i=k+1}^{n} \frac{k}{i-1} \approx \frac{k}{n} \log \frac{n}{k}.$$

Maximizing $x \log(1/x)$, $x = k/n$, gives $k \approx n/e$ and success probability $\to 1/e \approx 0.368$.

Example 7.8 (Uniform sums revisited)

Let $m(x) = E[N(x)]$, $N(x) = \min\{n : U_1 + \cdots + U_n > x\}$, $0 \leq x \leq 1$. Conditioning on $U_1 = y$ ($N = 1$ if $y > x$, else $1 + N'(x - y)$): $m(x) = 1 + \int_0^x m(u)\, du$, so $m' = m$, $m(0) = 1$, $m(x) = e^x$ (cf. Example 6.8).

Proposition 7.6 (Conditional variance formula)

With $\mathrm{Var}(X \mid Y) = E[(X - E[X \mid Y])^2 \mid Y]$,

$$\mathrm{Var}(X) = E\big[\mathrm{Var}(X \mid Y)\big] + \mathrm{Var}\big(E[X \mid Y]\big).$$

Proof. $E[\mathrm{Var}(X \mid Y)] = E[X^2] - E[(E[X \mid Y])^2]$ and $\mathrm{Var}(E[X \mid Y]) = E[(E[X \mid Y])^2] - (E[X])^2$; add. ∎

Example 7.9

(a) Random sum $S = \sum_1^N X_i$: $E[S \mid N] = N\mu$, $\mathrm{Var}(S \mid N) = N\sigma^2$, so $\mathrm{Var}(S) = E[N]\sigma^2 + \mu^2 \mathrm{Var}(N)$; for $N \sim \mathrm{Poisson}(\lambda)$, $\mathrm{Var}(S) = \lambda E[X^2]$.
(b) Geometric: with Y indicating first-trial success, $E[N \mid Y] \in \{1, 1 + 1/p\}$, $\mathrm{Var}(N \mid Y) \in \{0, \mathrm{Var}(N)\}$; solving, $\mathrm{Var}(N) = (1-p)/p^2$.

7.4 Conditional expectation and prediction

Motivation. $E[X \mid Y]$ minimizes mean squared prediction error; restricting to linear predictors gives regression.

Proposition 7.7

$E[(Y - g(X))^2] \geq E[(Y - E[Y \mid X])^2]$ for every function g: $E[Y \mid X]$ is the best predictor of Y from X in mean square error.

Proof. $E[(Y-c)^2 \mid X]$ is minimized by $c = E[Y \mid X]$; take expectations. ■

Minimizing $E[(Y - a - bX)^2]$ over linear predictors,

$$b = \rho \frac{\sigma_Y}{\sigma_X}, \qquad a = \mu_Y - b\mu_X, \qquad \mathrm{MSE} = \sigma_Y^2(1-\rho^2).$$

For the bivariate normal $E[Y \mid X]$ is linear. Galton's "regression to the mean" is the factor $|\rho| < 1$.

7.5 Moment generating functions

Motivation. The mgf turns independent sums into products and determines the distribution.

$M(t) = E[e^{tX}]$; when finite near 0, $M^{(n)}(0) = E[X^n]$.

Distribution	$M(t)$	valid for
$\mathrm{Bin}(n,p)$	$(pe^t + 1 - p)^n$	all t
$\mathrm{Poisson}(\lambda)$	$\exp\{\lambda(e^t - 1)\}$	all t
$\mathrm{Geometric}(p)$	$pe^t/\{1-(1-p)e^t\}$	$(1-p)e^t < 1$
$\mathrm{Exp}(\lambda)$	$\lambda/(\lambda - t)$	$t < \lambda$
$\mathrm{Gamma}(\alpha, \lambda)$	$\{\lambda/(\lambda-t)\}^\alpha$	$t < \lambda$
$N(\mu, \sigma^2)$	$\exp\{\mu t + \sigma^2 t^2/2\}$	all t

Normal: $E[e^{tZ}] = \frac{1}{\sqrt{2\pi}} \int e^{-(x-t)^2/2 + t^2/2}\, dx = e^{t^2/2}$, and $X = \mu + \sigma Z$.

Theorem 7.8

(a) If X, Y are independent, $M_{X+Y}(t) = M_X(t) M_Y(t)$.
(b) (Uniqueness) If $M_X = M_Y$ on a neighbourhood of 0, then X and Y have the same distribution.

(a) is Proposition 7.2 with $g = e^{tx}$, $h = e^{ty}$;

(b) is quoted. Together they give Proposition 6.2, e.g. $e^{\mu_1 t+\sigma_1^2 t^2/2}e^{\mu_2 t+\sigma_2^2 t^2/2}$ is the $N(\mu_1+\mu_2, \sigma_1^2+\sigma_2^2)$ mgf.

Example 7.10

(a) $E[e^{tZ^2}] = (1-2t)^{-1/2}$, so $\sum_1^n Z_i^2$ has the Gamma$(n/2, 1/2)$ mgf $(1-2t)^{-n/2}$.
(b) Random sums: $M_S(t) = E[M_X(t)^N]$; differentiating at 0 recovers Example 7.9.
(c) $M(t) = e^{3(e^t-1)}$: $X \sim$ Poisson(3), $P(X=0) = e^{-3}$.

The **joint mgf** $E[e^{t_1X_1+\cdots+t_nX_n}]$ determines the joint law, and factors iff the X_i are independent.

Example 7.11

For i.i.d. $N(\mu, \sigma^2)$ X, Y, $X+Y \perp X-Y$: $t(X+Y)+s(X-Y) = (t+s)X+(t-s)Y$, so

$$E[e^{t(X+Y)+s(X-Y)}] = e^{2\mu t+\sigma^2(t^2+s^2)} = E[e^{t(X+Y)}]\,E[e^{s(X-Y)}].$$

7.6 The multivariate normal distribution

If $Z_1, \ldots, Z_n$ are i.i.d. $N(0,1)$ and $X_i = \mu_i + \sum_j a_{ij}Z_j$, $(X_1, \ldots, X_m)$ is **multivariate normal**; every $\sum t_iX_i$ is normal with mean $\sum t_i\mu_i$ and variance $\sum_{i,j} t_it_j \operatorname{Cov}(X_i, X_j)$, so

$$M(t_1, \ldots, t_m) = \exp\Big\{\sum_i t_i\mu_i + \frac{1}{2}\sum_{i,j} t_it_j \operatorname{Cov}(X_i, X_j)\Big\}:$$

means and covariances determine the law, and uncorrelated components are independent.

Theorem 7.9 (Sample mean and variance)

If $X_1, \ldots, X_n$ are i.i.d. $N(\mu, \sigma^2)$, then $\bar{X}$ and S^2 are independent, $\bar{X} \sim N(\mu, \sigma^2/n)$, and $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$.

Proof. $(\bar{X}, X_1-\bar{X}, \ldots, X_n-\bar{X})$ is multivariate normal with $\operatorname{Cov}(\bar{X}, X_i-\bar{X}) = 0$, so $\bar{X}$ is independent of the deviations and of S^2. Next,

$$\sum_{i=1}^n \Big(\frac{X_i-\mu}{\sigma}\Big)^2 = \frac{(n-1)S^2}{\sigma^2} + \Big(\frac{\bar{X}-\mu}{\sigma/\sqrt{n}}\Big)^2.$$

The left side is χ^2_n, the last term χ^2_1 independent of the first; dividing mgfs, $(n-1)S^2/\sigma^2$ has mgf $(1-2t)^{-(n-1)/2}$. ∎

7.7 General definition of expectation

For general $X \geq 0$, $E[X] = \lim E[X_n]$ for simple $X_n \uparrow X$ (Lebesgue integral), or $\int_0^\infty P(X > x)\,dx$; $E[X] = E[X^+] - E[X^-]$ in general. All properties persist; e.g. if $X = Y$ w.p. p and W otherwise, $E[X] = pE[Y] + (1-p)E[W]$.

7.8 Selected exercises

Motivation. Ross's theoretical exercises (book numbering): indicators, conditioning, and the moments of a branching process.

Exercise 7.13 (Record values)

Let $X_1, \dots, X_n$ be i.i.d. continuous random variables. A record occurs at time j if $X_j \geq X_i$ for all $i \leq j$. Show that
(a) $E[\text{number of records}] = \sum_{j=1}^{n} 1/j$;
(b) $\text{Var}(\text{number of records}) = \sum_{j=1}^{n} (j-1)/j^2$.

Solution.

Step 1. (Ranks are uniform.) $P(X_i = X_k) = 0$ for $i \neq k$, and the joint density of $(X_1, \dots, X_m)$ is symmetric, so each of the $m!$ orderings has probability $1/m!$.

Step 2. I_j indicates $X_j = \max(X_1, \dots, X_j)$; $(j-1)!$ of $j!$ orderings put X_j on top, so $P(I_j = 1) = 1/j$, giving (a).

Step 3. For $i < j$, orderings with X_j on top correspond to the $(j-1)!$ orderings of $X_1, \dots, X_{j-1}$, a fraction $1/i$ of which puts X_i above $X_1, \dots, X_{i-1}$. So $P(I_i I_j = 1) = \frac{1}{ij} = P(I_i = 1)P(I_j = 1)$ and $\text{Cov}(I_i, I_j) = 0$.

Step 4. By Proposition 7.3,

$$\text{Var}\Big(\sum_{j=1}^{n} I_j\Big) = \sum_{j=1}^{n} \text{Var}(I_j) = \sum_{j=1}^{n} \frac{1}{j}\Big(1 - \frac{1}{j}\Big) = \sum_{j=1}^{n} \frac{j-1}{j^2}.$$

■

Exercise 7.20 (The conditional covariance formula)

For X, Y, Z with $E[X^2], E[Y^2] < \infty$ define $\text{Cov}(X, Y \mid Z) = E\big[(X - E[X \mid Z])(Y - E[Y \mid Z]) \mid Z\big]$.
(a) Show that $\text{Cov}(X, Y \mid Z) = E[XY \mid Z] - E[X \mid Z]E[Y \mid Z]$.
(b) Prove that $\text{Cov}(X, Y) = E[\text{Cov}(X, Y \mid Z)] + \text{Cov}(E[X \mid Z], E[Y \mid Z])$.
(c) Deduce the conditional variance formula.

Solution.

(a) $E[g(Z)V \mid Z] = g(Z)E[V \mid Z]$; expanding with $g(Z) = E[X \mid Z]$, $E[Y \mid Z]$,

$$\begin{aligned}\text{Cov}(X, Y \mid Z) &= E[XY \mid Z] - E[X \mid Z]E[Y \mid Z] - E[Y \mid Z]E[X \mid Z] + E[X \mid Z]E[Y \mid Z] \\ &= E[XY \mid Z] - E[X \mid Z]E[Y \mid Z].\end{aligned}$$

(b) Take expectations in (a) (Theorem 7.5):

$$E[\text{Cov}(X, Y \mid Z)] = E[XY] - E\big[E[X \mid Z]E[Y \mid Z]\big].$$

and, since $E\big[E[X \mid Z]\big] = E[X]$,

$$\text{Cov}(E[X \mid Z], E[Y \mid Z]) = E\big[E[X \mid Z]E[Y \mid Z]\big] - E[X]E[Y].$$

Add.

(c) $Y = X$ gives Proposition 7.6. ■

Exercise 7.35 (Generating functions as probabilities)

Let X be nonnegative integer valued with $p_j = P(X = j)$ and generating function $\phi(s) = E[s^X] = \sum_j p_j s^j$. For $0 < s < 1$ let Y be geometric with parameter $1 - s$, independent of X.

Show that $\phi(s) = P(X < Y)$.

Solution.

Step 1. $P(Y = k) = s^{k-1}(1-s)$, so $P(Y > j) = s^j$.

Step 2. Conditioning on X,

$$P(X < Y) = \sum_{j=0}^{\infty} P(X = j)P(Y > j) = \sum_{j=0}^{\infty} p_j s^j = \phi(s).$$

■

Exercise 7.44 (Moments of a branching process)

Each individual of a population produces j offspring with probability P_j, independently of all others; $\mu = \sum_j jP_j$ and $\sigma^2 = \sum_j (j-\mu)^2 P_j < \infty$. Let X_n be the size of generation n, with $X_0 = 1$. Show that
(a) $E[X_n] = \mu E[X_{n-1}]$, hence $E[X_n] = \mu^n$;
(b) $\operatorname{Var}(X_n) = \sigma^2\mu^{n-1} + \mu^2 \operatorname{Var}(X_{n-1})$;
(c) $\operatorname{Var}(X_n) = \sigma^2\mu^{n-1}\dfrac{\mu^n - 1}{\mu - 1}$ if $\mu \neq 1$, and $\operatorname{Var}(X_n) = n\sigma^2$ if $\mu = 1$.

Solution.

Step 1. $X_n = \sum_{i=1}^{X_{n-1}} Z_i$ with Z_i the offspring counts, i.i.d. (mean μ, variance σ^2) and independent of X_{n-1}:

$$E[X_n \mid X_{n-1}] = \mu X_{n-1}, \qquad \operatorname{Var}(X_n \mid X_{n-1}) = \sigma^2 X_{n-1}.$$

(a) Tower property and $E[X_0] = 1$.

(b) Proposition 7.6 and (a):

$$\operatorname{Var}(X_n) = E[\sigma^2 X_{n-1}] + \operatorname{Var}(\mu X_{n-1}) = \sigma^2\mu^{n-1} + \mu^2 \operatorname{Var}(X_{n-1}).$$

(c) $V_n = \operatorname{Var}(X_n)$, $V_0 = 0$. If $\mu = 1$, $V_n = \sigma^2 + V_{n-1} = n\sigma^2$. If $\mu \neq 1$, by induction

$$\begin{aligned} V_n &= \sigma^2\mu^{n-1} + \mu^2\sigma^2\mu^{n-2}\frac{\mu^{n-1}-1}{\mu-1} = \sigma^2\mu^{n-1}\,\frac{(\mu-1) + \mu(\mu^{n-1}-1)}{\mu-1} \\ &= \sigma^2\mu^{n-1}\frac{\mu^n - 1}{\mu - 1}. \end{aligned}$$

■